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# Effect of Temperature on Photovoltaic Cell Efficiency

Laboratory Report · Adam Nowak, Kasia Wiśniewska, Michał Kowalczyk · v14

## Abstract

This report examines how cell temperature affects the conversion efficiency of a monocrystalline silicon cell. Measurements were taken between 25 °C and 65 °C under constant illumination of 1000 W/m<sup>2</sup>.

## 1. Objective

Determine the temperature coefficient of efficiency for the test cell and compare it with values reported in the literature.

## 2. Theoretical Background

Photovoltaic efficiency decreases as temperature increases. **1** ~~Heat makes solar cells work worse.~~ The open-circuit voltage falls by roughly 2.2 mV per kelvin, while the short-circuit current rises only slightly **2** . As a result, the fill factor and the maximum power point both shift downward.

- Voc decreases by about 2.2 mV/K (lecture\_05.pdf, p. 22)
- Isc increases by about 0.05 %/K
- Net efficiency loss of about 0.41 %/K

### 3. Methodology

We put the cell on the plate and made it hotter step by step and wrote down the numbers.

## 4. Results

**Table 1 — Electrical parameters of the test cell at 1000 W/m<sup>2</sup>**

| T (°C) | Voc (V) | Isc (mA) | η (%) |
|--------|---------|----------|-------|
| 25     | 0.612   | 34.8     | 19.4  |
| 35     | 0.589   | 35.1     | 18.6  |

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